Gas-Phase and Heterogeneous Equilibria

Kp, Kc and equilibria with solids and liquids

Name _____ Class _____ Date _____

EQUATIONS AND DATA

Kp = Kc(RT)^Δn · Δn = gas moles (products − reactants) · R = 0.08206 L·atm/(mol·K) · T(K) = t(°C) + 273.15

Leave pure solids and liquids out of K · Q < K: forward · Q > K: reverse

Pressures in atm; K written without units

Count only gas moles for Δn. Show units for R and T.

A. Writing K

1. Write Kp for: (a) $2\,SO_2\,(g) + O_2\,(g) \rightleftharpoons 2\,SO_3\,(g)$ (b) $2\,NO\,(g) + O_2\,(g) \rightleftharpoons 2\,NO_2\,(g)$ (c) $H_2O\,(l) \rightleftharpoons H_2O\,(g)$ (d) $NH_4Cl\,(s) \rightleftharpoons NH_3\,(g) + HCl\,(g)$

2. Give Δn for each reaction in question 1, and say whether Kp is larger than, smaller than or equal to Kc at 25 °C.

B. Calculating Kp and Kc

3. At equilibrium, P(SO₂) = 0.20 atm, P(O₂) = 0.10 atm and P(SO₃) = 1.2 atm. Calculate Kp for reaction 1(a).

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4. For reaction 1(a), Kc = 280 at 1000 K. Calculate Kp.

5. For $N_2O_4\,(g) \rightleftharpoons 2\,NO_2\,(g)$, Kp = 0.15 at 25 °C. Calculate Kc.

C. Q and ICE tables

6. For $H_2\,(g) + I_2\,(g) \rightleftharpoons 2\,HI\,(g)$, Kp = 50 at a certain temperature. A mixture contains H_2 and I_2 at 0.10 atm each and HI at 0.50 atm. Calculate Q and predict the direction of change.

7. Pure N_2O_4 at 1.00 atm is allowed to reach equilibrium at 25 °C (Kp = 0.15). Calculate the equilibrium partial pressures of N_2O_4 and NO_2.

D. Challenge

8. Solid $CaCO_3$ is heated in a sealed container until $CaCO_3\,(s) \rightleftharpoons CaO\,(s) + CO_2\,(g)$ reaches equilibrium. Explain what happens to the pressure of CO_2 if (a) more $CaCO_3$ is added (b) some CO_2 is pumped out (at constant temperature).

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